# NUMBER PATTERNS

Short Revision Notes · Grade 9 Mathematics

**Understand Level — Explain Ideas & Concepts**

## 1. WHAT IS A NUMBER PATTERN?

A number pattern (or sequence) is an **ordered list of numbers** that follow a specific rule. The **order matters!**

| Key Concept | Explanation |
|---|---|
| Sequence vs Pattern | In mathematics, "sequence" and "pattern" mean the same thing |
| Order Matters | 1, 1, 2, 2, 3, 3 is DIFFERENT from 1, 2, 1, 2, 3, 4 |
| Ellipsis (…) | The three dots at the end mean "the pattern continues" |
| Rule Needed | To describe a sequence accurately, you need a rule to calculate each term |

## 2. WHY THE FIRST FEW TERMS ARE NOT ENOUGH

| Concept | Explanation | Example |
|---|---|---|
| Ambiguity | Just looking at first few terms is mathematically incorrect | If you see 1, 2, 3, 4, 5 … you might think 6 is next, but the pattern could be 1, 2, 3, 4, 5, 1, 2, 3, 4, 5, … |
| Multiple Patterns | Many different rules can fit the same first few terms | The first five terms could follow many different patterns |
| Need a Rule | A proper sequence must have a clear rule explaining how terms are generated | Without a rule, the sequence is not properly defined |

## 3. THE COMMON DIFFERENCE CONCEPT

| Concept | Explanation | Example |
|---|---|---|
| Definition | The constant value added to each term to get the next term | In 2, 4, 6, 8, … the common difference is 2 |
| How to Find It | Subtract any term (except the first) from the previous term | 8 – 6 = 2 |
| Can Be Negative | If terms are decreasing, the common difference is negative | 17, 14, 11, 8 has common difference –3 |
| Can Be Zero | If all terms are the same, the common difference is 0 | 3, 3, 3, 3, … has common difference 0 |
| "Common" Meaning | "Common" means it's the same for the entire sequence | The difference doesn't change throughout the sequence |

## 4. THE GENERAL TERM CONCEPT

| Concept | Explanation | Example |
|---|---|---|
| What It Is | An expression that gives any term when you substitute n | $T_n = 2n + 3$ |
| Why It's Important | It allows you to find ANY term (even the 1000th!) without writing all previous terms | $T_{1000} = 2(1000) + 3 = 2003$ |
| What n Represents | The position of the term (1st, 2nd, 3rd…) | For the 20th term, n = 20 |
| What $T_n$ Represents | The value of the term at position n | $T_1 = 5$ means the 1st term is 5 |

## 5. RELATING COMMON DIFFERENCE TO THE GENERAL TERM

| Concept | Explanation | Example |
|---|---|---|
| Coefficient of n | The common difference is ALWAYS the coefficient of n in the general term | $T_n = 4n + 5 \rightarrow$ common difference = 4 |
| Why This Works | Each time you increase n by 1, you add the common difference | From $T_1$ to $T_2$, n increases by 1 → add common difference |
| Constant Term | The number added to the coefficient × n adjusts the first term | $T_n = 4n + 5 \rightarrow T_1 = 4(1) + 5 = 9$ |

## 6. HOW TO DEVELOP THE GENERAL TERM

| Step | Explanation | Example: 5, 11, 17, 23, … |
|---|---|---|
| 1 | Find the common difference | 11 – 5 = 6 |
| 2 | Write multiples of the common difference | 6, 12, 18, 24, … |
| 3 | Compare sequence to the multiples | 5, 11, 17, 23 vs 6, 12, 18, 24 |
| 4 | See what must be added/subtracted | Each term is 1 less than the multiple |
| 5 | Write the general term | $T_n = 6n - 1$ |

**Alternative method (formula approach):**

$$T_n = \text{First term} + (n - 1) \times \text{common difference}$$

$$T_n = 5 + (n - 1) \times 6$$

$$T_n = 5 + 6n - 6$$

$$T_n = 6n - 1$$

## 7. UNDERSTANDING DIFFERENT TYPES OF SEQUENCES

| Type | Description | Example | Common Diff? |
|---|---|---|---|
| **Same Difference Sequence** | Constant value is added each time | 2, 4, 6, 8, 10, … | Yes ✓ |
| **Decreasing Sequence** | Negative value is added (subtracting) | 10, 7, 4, 1, –2, … | Yes ✓ |
| **All Same Sequence** | Zero is added each time | 3, 3, 3, 3, … | Yes ✓ |
| **Multiplication Sequence** | Terms are multiplied by a constant | 2, 4, 8, 16, 32, … | No ✗ |
| **Alternating Sequence** | Different values repeat in a pattern | 1, 1, 2, 2, 3, 3, … | No ✗ |

■ **Key Understanding:** ONLY sequences where you **ADD the same number each time** are covered by this chapter!

## 8. UNDERSTANDING THE (n+1)th TERM

| Concept | Explanation | Example |
|---|---|---|
| What It Is | The term that comes AFTER the n-th term | If n = 3, the (n+1)th term is the 4th term |
| How to Find It | Substitute (n+1) for n in the general term | $T_n = 2n + 3 \rightarrow T_{n+1} = 2(n+1) + 3$ |
| Simplify | Expand and simplify the expression | $T_{n+1} = 2n + 2 + 3 = 2n + 5$ |
| Why It's Useful | It shows the relationship between consecutive terms | Each term is common difference more than the previous |

## 9. UNDERSTANDING "SHOW THAT" PROBLEMS

| Type | What It Means | Example |
|---|---|---|
| "Show that 0 is a term" | Find an integer n such that $T_n = 0$ | If $T_n = 56 - 4n$, set $56 - 4n = 0 \rightarrow n = 14$ |
| "Show that 18 is NOT a term" | Show that n is NOT a whole number | If $T_n = 56 - 4n = 18 \rightarrow n = 9.5$ (not whole) |
| "Show terms after 7th are negative" | Find when $T_n < 0$ | $T_n = 50 - 7n \rightarrow 50 - 7n < 0 \rightarrow n > 7.14 \rightarrow n \geq 8$ |

## 10. UNDERSTANDING THE FORMULA METHOD

| Formula | Explanation | When to Use |
|---|---|---|
| $T_n$ = First term + (n–1) × d | Direct formula using first term and common difference | When you know the first term and common difference |
| $T_n = d \times n + c$ | General form where d is common difference and c is constant | When you've found d and need to find c |
| Finding c | $T_1 - d = c$ | Substitute the first term and common difference |

**Understanding Check:**

| If the common difference is | And the first term is | Then the general term is |
|---|---|---|
| +3 | 5 | $T_n = 3n + 2$ |
| +7 | 10 | $T_n = 7n + 3$ |
| –4 | 7 | $T_n = 11 – 4n$ |
| +6 | 15 | $T_n = 6n + 9$ |

## 11. QUICK UNDERSTANDING CHECK

| Concept | Your Understanding |
|---|---|
| What is a sequence? | A pattern of numbers with a rule |
| Why are first few terms ambiguous? | Many different rules could fit |
| What is common difference? | The constant value added each time |
| What does n mean in $T_n$? | The position of the term |
| What is the general term? | An expression to find any term |
| How is common difference related to $T_n$? | It's the coefficient of n |
| What is the (n+1)th term? | The term after the n-th term |
| Why use general term? | To find any term quickly |

## 12. SELF-TEST — UNDERSTANDING QUESTIONS

| No. | Question | Answer |
|---|---|---|
| 1 | Why can't you uniquely determine a sequence from just 5 terms? | **Many different patterns could fit the same first 5 terms** |
| 2 | What does "common" in "common difference" mean? | **The value remains the same throughout the sequence** |
| 3 | If $T_n = 4n + 7$, what is the common difference? | **4** |
| 4 | What does the coefficient of n always represent? | **The common difference** |
| 5 | How is $T_{n+1}$ related to $T_n$? | **$T_{n+1} = T_n$ + common difference** |
| 6 | Why would a sequence with common difference 0 be all the same number? | **Because adding 0 each time doesn't change the value** |
| 7 | What does it mean if the common difference is negative? | **The sequence is decreasing** |
| 8 | Why is the general term better than listing terms? | **You can find any term without writing all previous terms** |
| 9 | What is the difference between a sequence and just a list of numbers? | **A sequence follows a specific rule** |

| No. | Question | Answer |
| --- | --- | --- |
| 10 | If $T_n = 5 - 2n$, what does the 5 represent? | **When n=0, the value would be 5 (the starting point before the first term)** |

## 13. KEY UNDERSTANDINGS TO REMEMBER

■ **A sequence is DEFINED by its rule**, not just its first few terms.

■ **The common difference is CONSTANT** for the entire sequence.

■ **The coefficient of n** in $T_n$ is **ALWAYS the common difference**.

■ **The general term is a POWERFUL tool** — it can find ANY term!

■ **When a term "does not exist"**, n will NOT be a whole number.

■ For a sequence with common difference d and first term a:

$T_n = a + (n - 1)d = dn + (a - d)$

★ **Revision Tip:** Practice explaining these concepts to a friend. If you can **teach it**, you truly understand it!